

Unit 12, Lesson 5: Writing Descriptions of Real-World Contexts from Algebraic Expressions Equation

Benchmark

MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.

Learning Target

- ☐ I can write a description of a real-world context from an algebraic expression.

12.5.1 Warm-Up: Self-Made Millionaire

Madam C.J. Walker was the first African American self-made millionaire. She made her fortune by inventing and selling specialized hair products, including an innovative Edge Control product that sold for \$0.35 each beginning in 1905.



1. Create an algebraic expression to describe the total revenue Madam C.J. Walker earned for selling any number of Edge Control products, e .

12.5.2 Activity: Matching Real-World Contexts to Expressions

1. Match each real-world situation with the correct expression.

A. Ryan has 20 songs. His friend bought him x more songs. How many songs does Ryan have?

B. Karen has x flowers. Helen has 20 times more flowers than Karen. How many flowers does Helen have?

C. George earned 20 points in a card game. Bruno earned x fewer points than George. How many points did Bruno earn?

D. Maria has 20 books and wants to put an equal number of books on x bookshelves. How many books go on each shelf?

1. $20x$

2. $x + 20$

3. $\frac{20}{x}$

4. $20 - x$

12.5.3 Collaboration: Writing Real-World Contexts

1. For each expression, create a real-world scenario that the algebraic expression could represent.

Expression	Your Real-World Scenario	What does the variable represent?
$r - 10$		
$21y$		
$\frac{100}{a}$		
$17 + s$		
$57 - r$		
$48w$		

2. With your partner, pick four expressions from Question 1 and use them to complete the following table.

Expression	Your Partner's Scenario	Does the Expression Accurately Represent the Scenario?	Similarities Between Your and Your Partner's Scenarios for the Expression
		☐ Yes ☐ No	
		☐ Yes ☐ No	
		☐ Yes ☐ No	
		☐ Yes ☐ No	

12.5.4 Activity: Gallery Walk

Several scenarios are displayed. For each scenario, write an algebraic expression to match.

Scenario	Expression

12.5.5 Wrap-Up: Ticket Out the Door

1. Frank has a plant that measured h inches tall last week. This week the plant is 8 inches tall.
 - a. Circle the expression that represents the number of inches, h , the plant grew in a week.

$8 - h$ $h - 8$
 - b. Explain your choice.
 - c. For the expression not chosen, describe a real-world situation that the expression might represent.

